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Winter 2025

B.Tech 4th Semester Examination

Surveying & Geomatics

Course Code: BCE23402T

Time – 2 hours

Full Marks – 50

The figure in the margin indicates full marks for the questions.

SECTION 1

(10x1=10)

I. Choose the correct option:

1. Why is leveling performed in surveying projects?
A) to calculate the area of land
B) to determine the height of structures
C) to find the relative heights of different points
D) to align boundaries

2. Which of the following best explains the purpose of a theodolite in surveying?
A) It is used to measure distances with high accuracy
B) It is used to set out curves on highways
C) It is used to measure horizontal and vertical angles
D) It is used to draw survey maps

3. In plane table surveying, what is meant by orientation?
A) Setting up the instrument on a tripod
B) aligning the drawing board to the magnetic north or a known direction
C) Measuring angles between features
D) marking station points on the table

4. Which of the following best describes a contour line?
A) a line showing property boundaries
B) a line representing constant magnetic variation
C) a line connecting points of equal elevation
D) a line used for compass bearings

5. In aerial photogrammetric, photographs are taken from:
A. Satellites
B. Ground stations
C. Aircrafts or drones
D. Weather balloons

6. What is a Total Station?
A. a type of leveling instrument
B. a digital theodolite with an electronic distance measurement (EDM) device
C. a mechanical compass
D. a GPS receiver

7. What is the purpose of curve setting in civil engineering?
A. To increase the height of a structure
B. To connect two straight lines smoothly
C. To design the drainage system
D. To determine soil strength

8. In the prismoidal formula, the volume is calculated as:
A. (A1 + A2) × L
B. (A1 + 4Am + A2) × L / 6
C. (A1 + A2 + Am) / 3
D. A1 × A2 × L

9. What defines a well-conditioned triangle in triangulation?
A. A triangle where all angles are less than 30°
B. A triangle with one angle more than 120°
C. A triangle with all angles nearly equal (close to 60°)
D. A triangle with sides in the ratio 1:2:3

10. Local attraction in compass surveying is caused by:
A. Magnetic declination
B. Variation in temperature
C. Presence of magnetic substances nearby
D. Errors in chain measurements

SECTION 2

II. Answer any one question from this section:

2.

a) Define a compound curve. Two tangents intersect at a chainage of 1350 m. The angle of intersection is 160°. Calculate all data necessary for setting out a curve of radius 260.00 m. (2+8=10)

b) What is the relation between degree and radius of a curve? Explain what you understand by degree of a curve. Define the apex and deflection angle of a simple circular curve with proper diagram. (2+2+3+3=10)

SECTION 3

III. Answer any one question from this section:

3.

a) State the advantages and disadvantages of plane table surveying. Explain what you understand by orientation of a plane table. (10)

b) Convert WCB = 73° to QB, WCB = 200° to QB, Convert WCB = 0° to QB, S25°E to WCB, N70°W to WCB. State the differences between a prismatic compass and surveyor's compass. (5+5=10)

SECTION 4

IV. Answer any one question from this section:

4.

a) For the following given data, calculate the number of photographs required to represent and area of 5 km × 3 km (5000 m × 3000 m).Camera specifications: Focal length (f) = 152 mm. Image format = 230 mm × 230 mm. The Ground parameters are as follows: Flying height: 3040m. Required longitudinal overlap = 60%. Required lateral overlap=30%. (10)

b) A steel tape exactly 30.00 m long at 20 °C was used to measure a line. The pull applied on the tape during measurement was 15kgs while the standard pull was 10kgs.The field temperature during measurement was 45°C.The length of the line measured was found to be 790m long. If the cross sectional area of the tape = $0.03 cm^2$, its total weight= 0.693kgs and Coefficient of thermal expansion (α) = 11.2×10^{-6} /°C, Modulus of elasticity (E) = 2×10^5 N/mm², Find the corrected length of the line. (10)

SECTION 5

V. Answer any one question from this section:

5.

a) The following consecutive readings were taken with a level and 4.0 m leveling staff on a continuously sloping ground at common interval of 30.0m. The readings are as follows:
0.905, 1.745, 2.345, 3.125, 3.725, 0.545, 1.390, 2.055, 2.955, 3.455, 0.595, 1.015, 1.850, 2.655, and 2.945.The RL of A was 395.500. Calculate the RLs of different points. (10)

b. State the trapezoidal rule and Simpson's rule. Derive an expression for computing the area and volume of an embankment in cutting/ filling. (10)

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